

More information

Modelfile Reference

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A Modelfile is the blueprint to create and share customized models using Ollama.

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Format

The format of the `Modelfile` :

```
# comment
INSTRUCTION arguments
```

Instruction	Description
`FROM` (required)	Defines the base model to use.
`PARAMETER`	Sets the parameters for how Ollama will run the model.
`TEMPLATE`	The full prompt template to be sent to the model.
`SYSTEM`	Specifies the system message that will be set in the template.
`LICENSE`	Specifies the legal license.
`MESSAGE`	Specify message history.
`REQUIRES`	Specify the minimum version of Ollama required by the model.
`CAPABILITY`	Declare an additional model capability.

Examples

Basic `Modelfile`

An example of a `Modelfile` creating a mario blueprint:

```
FROM llama3.2
# sets the temperature to 1 [higher is more creative, lower is more coher
PARAMETER temperature 1
# sets the context window size to 4096, this controls how many tokens the
PARAMETER num_ctx 4096

# sets a custom system message to specify the behavior of the chat assist
SYSTEM You are Mario from super mario bros, acting as an assistant.
```



To use this:

1. Save it as a file (e.g. `Modelfile`)
2. `ollama create choose-a-model-name -f <location of the file e.g. ./Modelfile>`
3. `ollama run choose-a-model-name`
4. Start using the model!

To view the Modelfile of a given model, use the `ollama show --modelfile` command.

```
ollama show --modelfile llama3.2
```

```
# Modelfile generated by "ollama show"
# To build a new Modelfile based on this one, replace the FROM line with:
# FROM llama3.2:latest
FROM /Users/pdevine/.ollama/models/blobs/sha256-00e1317cbf74d901080d7100f
TEMPLATE """{{ if .System }}<|start_header_id|>system<|end_header_id|>

{{ .System }}<|eot_id|>{{ end }}{{ if .Prompt }}<|start_header_id|>user<|

{{ .Prompt }}<|eot_id|>{{ end }}<|start_header_id|>assistant<|end_header_

{{ .Response }}<|eot_id|>"""
PARAMETER stop "<|start_header_id|>"
PARAMETER stop "<|end_header_id|>"
PARAMETER stop "<|eot_id|>"
PARAMETER stop "<|reserved_special_token"
```

Instructions

FROM (Required)

The `FROM` instruction defines the base model to use when creating a model.

```
FROM <model name>:<tag>
```

Build from existing model

```
FROM llama3.2
```

> **Model library**
> Browse available models

Build from a Safetensors model

```
FROM <model directory>
```

The model directory should contain the Safetensors weights for a supported architecture.

Build from a GGUF model

```
FROM ./ollama-model.gguf
```

The GGUF file location should be specified as an absolute path or relative to the `Modelfile` location.

For a split GGUF model, keep the original split filenames and use a wildcard that matches every shard. Shards can also be listed with separate `FROM` instructions.

```
FROM ./model-*.gguf
```

PARAMETER



The `PARAMETER` instruction defines a parameter that can be set when the model is run.

```
PARAMETER <parameter> <parametervalue>
```

Valid Parameters and Values

Parameter	Description	Value Type	Example Usage
num_ctx	Sets the size of the context window used to generate the next token. (Default: 2048)	int	num_ctx 4096
repeat_last_n	Sets how far back for the model to look back to prevent repetition. (Default: 64, 0 = disabled, -1 = num_ctx)	int	repeat_last_n 64
repeat_penalty	Sets how strongly to penalize repetitions. A higher value (e.g., 1.5) will penalize repetitions more strongly, while a lower value (e.g., 0.9) will be more lenient. (Default: 1.0, disabled)	float	repeat_penalty 1.1
temperature	The temperature of the model. Increasing the temperature will make the model answer more creatively. (Default: 0.8)	float	temperature 0.7
seed	Sets the random number seed to use for generation. Setting this to a specific number will	int	seed 42

Parameter	Description	Value Type	Example Usage
	make the model generate the same text for the same prompt. (Default: 0)		
stop	Sets the stop sequences to use. When this pattern is encountered the LLM will stop generating text and return. Multiple stop patterns may be set by specifying multiple separate `stop` parameters in a modelfile.	string	stop “AI assistant:“
num_predict	Maximum number of tokens to predict when generating text. (Default: -1, infinite generation)	int	num_predict 42
draft_num_predict	Maximum number of speculative draft tokens to predict per step when a draft model is available. Separate draft models default to 4; embedded MTP tensors require setting this parameter. Set to 0 to disable speculative drafting.	int	draft_num_predict 4

Parameter	Description	Value Type	Example Usage
top_k	Reduces the probability of generating nonsense. A higher value (e.g. 100) will give more diverse answers, while a lower value (e.g. 10) will be more conservative. (Default: 40)	int	top_k 40
top_p	Works together with top-k. A higher value (e.g., 0.95) will lead to more diverse text, while a lower value (e.g., 0.5) will generate more focused and conservative text. (Default: 0.9)	float	top_p 0.9
min_p	Alternative to the top*p, and aims to ensure a balance of quality and variety. The parameter _p* represents the minimum probability for a token to be considered, relative to the probability of the most likely token. For example, with *p*=0.05 and the most likely token having a probability of 0.9, logits with a value less than 0.045 are filtered out. (Default: 0.0)	float	min_p 0.05

TEMPLATE



`TEMPLATE` of the full prompt template to be passed into the model. It may include (optionally) a system message, a user's message and the response from the model. Note: syntax may be model specific. Templates use Go **template syntax**.

Template Variables

Variable	Description
`{{ .System }}`	The system message used to specify custom behavior.
`{{ .Prompt }}`	The user prompt message.
`{{ .Response }}`	The response from the model. When generating a response, text after this variable is omitted.

```
TEMPLATE """{{ if .System }}<|im_start|>system
{{ .System }}<|im_end|>
{{ end }}{{ if .Prompt }}<|im_start|>user
{{ .Prompt }}<|im_end|>
{{ end }}<|im_start|>assistant
"""
```

SYSTEM

The `SYSTEM` instruction specifies the system message to be used in the template, if applicable.

```
SYSTEM """<system message>"""
```

LICENSE

The `LICENSE` instruction allows you to specify the legal license under which the model used with this Modelfile is shared or distributed.

```
LICENSE """
<license text>
"""
```

MESSAGE

The `MESSAGE` instruction allows you to specify a message history for the model to use when responding. Use multiple iterations of the MESSAGE command to build up a conversation which will guide the model to answer in a similar way.

```
MESSAGE <role> <message>
```

Valid roles

Role	Description
system	Alternate way of providing the SYSTEM message for the model.
user	An example message of what the user could have asked.
assistant	An example message of how the model should respond.

Example conversation

```
MESSAGE user Is Toronto in Canada?
MESSAGE assistant yes
MESSAGE user Is Sacramento in Canada?
MESSAGE assistant no
MESSAGE user Is Ontario in Canada?
MESSAGE assistant yes
```

CAPABILITY

Adds a capability to a GGUF or MLX model without removing inherited or inferred capabilities. Repeat the instruction to add more than one. Declare

`decision` only for weights trained for System One's decision prompts:

```
CAPABILITY decision
```

REQUIRES

The `REQUIRES` instruction allows you to specify the minimum version of Ollama required by the model.

```
REQUIRES <version>
```

The version should be a valid Ollama version (e.g. 0.14.0).

Notes

- the `Modelfile` **is not case sensitive**. In the examples, uppercase instructions are used to make it easier to distinguish it from arguments.
- Instructions can be in any order. In the examples, the `FROM` instruction is first to keep it easily readable.

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